Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



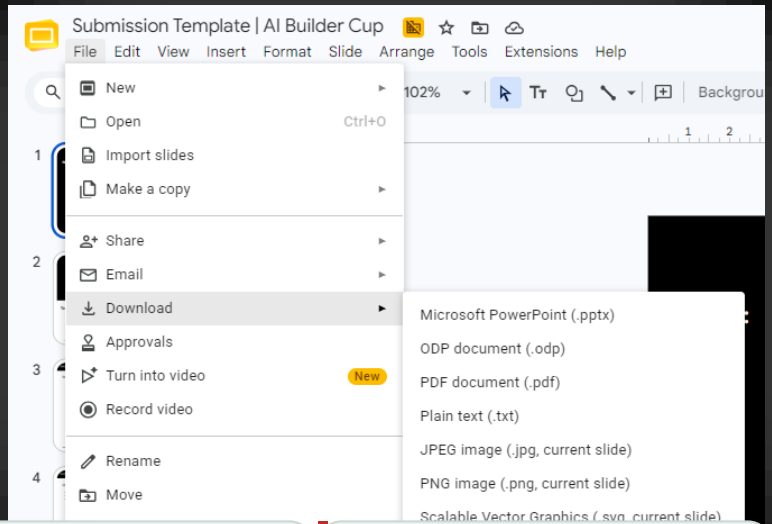
TEAM DETAILS · AI BUILDER CUP 2026 · JAPAC · SUSTAINABILITY

Team name: Agent Ram

Team leader: Ramasamy P

Product: TerraSentinel AI — ground-motion early warning for roads, metros & dams

Problem statement: After each Sentinel-1 pass, operators still face thousands of InSAR points with no consequence ranking. TerraSentinel cuts motion into 250 m sites, ranks who inspects today (RPI × tier), reports honest time-to-threshold bands (never failure dates), and uses Gemini to draft a number-verified inspection ticket.



TERRASENTINEL AI

Satellites see the ground move. Gemini tells engineers where to go first.

87 → 7
sites to the few that need a crew

RPI 86
NH-58 CRITICAL · 35–64 day band

Gemini
ticket from verified tool numbers

GCP
Cloud Run · BigQuery · Vertex AI

BRIEF ABOUT THE IDEA

Ground creeps for months. Nobody ranks which creep matters — so crews go to the wrong place.

Mountain highways · NH-58

Sole valley access. One slope slip cuts towns, pilgrims, supply.

Metro tunnels · Chennai

Settlement under roads & buildings. Tight tolerances, dense exposure.

Dams & tailings · mining belt

Crest settlement / toe motion with severe downstream consequence.

The one decision we support

After this pass: which 250 m needs an engineer, why, and how soon?

Today's options fail the decision

- Raw InSAR portal — thousands of points, no asset link, no "who goes today"
- Rainfall alarm — treats the trigger as the hazard; fires over stable ground
- Generic AI chatbot — fluent text that invents mm and failure dates
- Scheduled walk-over — calendar ignores which segment is accelerating now

We do

- Progressive creep on linear assets
- Rank, explain, track between passes
- Threshold-crossing window + assumptions

We refuse

- Failure dates / collapse prediction
- Sudden events between 6–12 day passes
- Certified safety decisions

OPPORTUNITIES · DIFFERENTIATION · USP

Not another InSAR viewer — a decision layer with honest outlooks and guarded GenAI.

How different?

- 250 m chainage sites on corridor geometry — not a heat-map of every scatterer
- RPI = hazard × consequence × confidence (deterministic Python / BigQuery)
- Pass-to-pass alerts = diffs; Act list = today's state
- Rain amplifies only (capped +50%); USGS quakes never score
- Gemini narrates tool results; engine owns every number

How it solves it

- Ingest → validate → locate on assets → measure speed/accel/progress
- Classify with WEAK data guard (weak radar can never be CRITICAL)
- Rank Act-today shortlist for engineers
- Outlook = [ETA_const, ETA_accel] band ≤ 365 d — never a date
- Work order: summary · hypothesis · urgency · field checklist

USP

- Honest ETA bands with continuing-regime assumption spelled out
- Number guard rejects invented mm / "will collapse" language
- Same maths local CSV XOR BigQuery (TS_DATA_BACKEND)
- Production GCP posture: Cloud Run, Secret Manager, GCS state
- Demo value: 62,330 obs → 87 sites → 7 action → 4 Act today

FEATURES OFFERED BY THE SOLUTION

Command Center to Work Order — every surface ships a decision, not a dashboard toy.

Command Center

Map + KPIs + Act today. CRITICAL/ELEVATED ranked by consequence, not raw mm.

Site drawer

RPI, speed vs baseline, rain trigger, outlook band, trend across passes.

Alerts

What changed between satellite passes — not a duplicate of the Act list.

Brief · Early Warning · Trends · Portfolio

Planning views; history recomputed from data so past scores reproduce.

Work Orders + Gemini

Template-first ticket; Gemini polishes via 7 tools; guard verifies numbers.

Data Hub

Validate-then-import CSV/GeoJSON; Open-Meteo rain; provenance; BigQuery sync.

Access & audit

admin / analyst / viewer RBAC · sessions · CSRF · audit log.

Methodology

Published formulae so every score is auditable by an engineer.

PROCESS / USE-CASE FLOW

One satellite pass: from space to a crew at the right 250 metres.

1 · Pass
Every 6–12 days

Sentinel-1 measures mm motion through cloud & night

2 · Map
Onto the asset

1,355 points → highway / metro / dam → 87×250 m sites

3 · Rank
What matters

Fast + accelerating + critical + good data rises; rain alone never alarms

4 · Gemini
Ticket

Tools → reason → JSON → number guard → checklist

5 · Next
Keep watching

Crew acts; next pass shows calm-down or worse

One pass on the demo dataset (labelled synthetic)

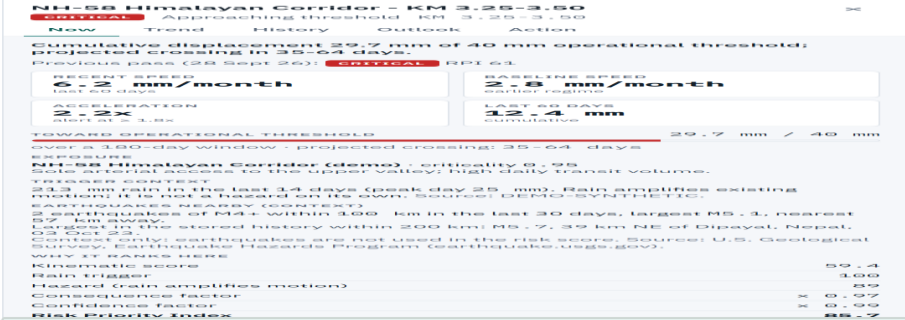
62,330	1,355	87	7	4
disp. obs	radar pts	250 m sites	CRIT/ELEV	Act today

PRODUCT UI (LIVE PROTOTYPE)

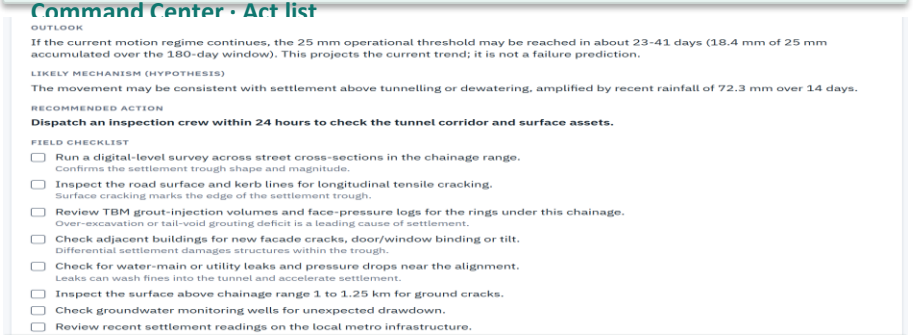
Not wireframes — the running product judges can open today.



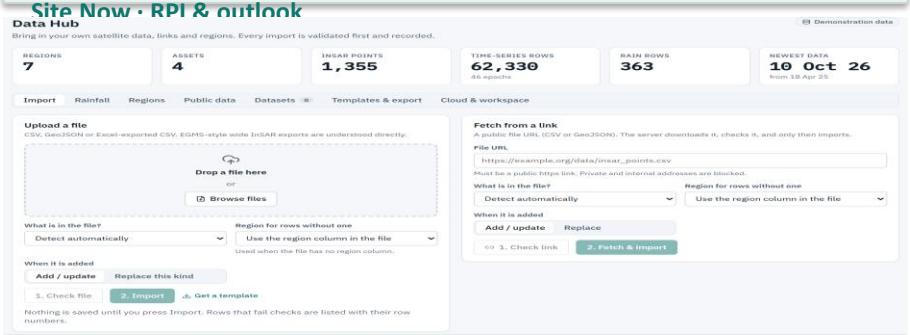
Command Center · Act list



Site Now · RPI & outlook



Gemini work order



Data Hub · validate→import

ARCHITECTURE ON GOOGLE CLOUD

One container · two storage backends · the same tested maths.

EDGE	Cloud Run · HTTPS · CSP/HSTS · rate limits · session cookie · GCS state volume
UI	React 19 · MapLibre · React Query · role-aware screens
API	FastAPI · RBAC viewer/analyst/admin · CSRF · SSRF-safe fetch · audit
ENGINE	numpy · pandas · shapely — metrics, RPI, tiers, ETA bands, alerts, pass replay
GENAI	Gemini tool loop · number guard · model fallback · AI Studio or Vertex (ADC)
DATA	BigQuery (GEOGRAPHY, partitions) XOR local · USGS store · provenance

TECHNOLOGIES

Google Cloud end-to-end. Gemini turns a risk score into an action.

Ingest	Store	Compute	Explain	Act
Cloud Run	BigQuery	Cloud Run	Gemini	Engineers
Data Hub validate	GIS join · history	risk engine	Vertex / Studio	map · ticket · audit

Gemini (what GenAI does)

- 7 read-only tools: metrics, history, outlook, rain, seismic, exposure, quality
- Round 0 MUST call tools → investigate → write JSON → guard
- Rejects inventing mm/days; rejects “will collapse” / failure dates
- Temp 0.1 · budget · fallback flash-latest → lite on 429/5xx
- Live proof: 6 tool calls · 7 numbers verified · “inspect within 24h”

GCP services & why

- Vertex AI / AI Studio — Gemini for production & dev
- BigQuery — 9 tables, partitions, ST_DWITHIN join, bytes-billed cap
- Cloud Run — one image: SPA + API + engine + agent
- Secret Manager · GCS state · IAM via ADC (no key files)
- Open-Meteo + USGS egress through SSRF-safe fetcher

ESTIMATED IMPLEMENTATION COST (OPTIONAL)

Cup-scale economics: GenAI only on tickets; ranking stays deterministic.

Cloud Run	1–2 vCPU / 2 GiB · min-instances=1 while judging · scales to 0 off-hours	**low tens USD/mo warm**
BigQuery	~62k TS rows · in-process TTL cache · maximum_bytes_billed on every job	**pennies–low USD demos**
Gemini Flash	Work-order path only · template fallback · not on hot ranking path	**per-ticket, optional**
Secrets + GCS	Admin/Gemini secrets · state volume for sessions/audit	**pennies**
Scale path	Materialize assignment · raise instances · keep engine CPU-bound	**linear with corridors**

SNAPSHOTS OF THE PROTOTYPE

Demo value on screen — Chamoli corridors, NH-58 CRITICAL, guarded ticket.





NH-58 Himalayan Corridor – KM 3.25–3.50
CRITICAL · Approaching threshold · KM 3.25 – 3.50
Now · Trend · History · Outlook · Action

Cumulative displacement 29.7 mm of 40 mm operational threshold; projected crossing in 35–64 days.
Previous pass (28 Sept 26): CRITICAL · RPI 61.

RECENT SPEED	BASELINE SPEED
6.3 mm/month — last 60 days	2.8 mm/month — earlier regime
ACCELERATION	LAST 60 DAYS
2.23× — alert at ≥ 1.8×	12.4 mm — cumulative

TOWARD OPERATIONAL THRESHOLD — 29.7 mm / 40 mm
over a 180-day window · projected crossing: 35–64 days

EXPOSURE
NH-58 Himalayan Corridor (demo) · criticality 0.95
Sole arterial access to the upper valley; high daily transit volume.

TRIGGER CONTEXT
213 mm rain in the last 14 days (peak day 25 mm); Rain amplifies existing movement (it is not a hazard on its own). Source: DEMO-SYNTHETIC.

EARTHQUAKES NEARBY (CONTEXT)
2 earthquakes of M4.4+ within 300 km in the last 30 days, largest M5.1, nearest 57 km away.
Largest in the stored history within 200 km: M5.7, 39 km NE of Dipayal, Nepal, 03 Oct 23.
Context only; earthquakes are not used in the risk score. Source: U.S. Geological Survey, Earthquake Hazards Program (earthquake.usgs.gov).

WHY IT READS HERE

Kinematic score	59.4
Rain trigger	1.00
Hazard (rain amplifies motion)	.89
Consequence factor	× 0.97
Confidence factor	× 0.99
Risk Priority Index	**25.7**

OUTLOOK
If the current motion regime continues, the 25 mm operational threshold may be reached in about 23–41 days (18.4 mm of 25 mm accumulated over the 180-day window). This projects the current trend; it is not a failure prediction.

LIKELY MECHANISM (HYPOTHESIS)
The movement may be consistent with settlement above tunnelling or dewatering, amplified by recent rainfall of 72.3 mm over 14 days.

RECOMMENDED ACTION
Dispatch an inspection crew within 24 hours to check the tunnel corridor and surface assets.

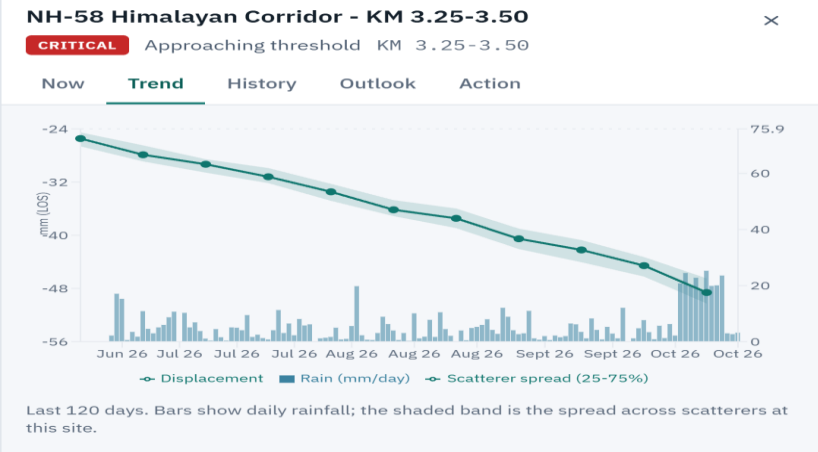
FIELD CHECKLIST
- [ ] Run a digital-level survey across street cross-sections in the chainage range.
  Confirms the settlement trough shape and magnitude.
- [ ] Inspect the road surface and kerb lines for longitudinal tensile cracking.
  Surface cracking marks the edge of the settlement trough.
- [ ] Review TBM grout-injection volumes and face-pressure logs for the rings under this chainage.
  Over-excavation or tail-void grouting deficit is a leading cause of settlement.
- [ ] Check adjacent buildings for new facade cracks, door/window binding or tilt.
  Differential settlement damages structures within the trough.
- [ ] Check for water-main or utility leaks and pressure drops near the alignment.
  Leaks can wash fines into the tunnel and accelerate settlement.
- [ ] Inspect the surface above chainage range 1 to 1.25 km for ground cracks.
- [ ] Check groundwater monitoring wells for unexpected drawdown.
- [ ] Review recent settlement readings on the local metro infrastructure.

PROTOTYPE PERFORMANCE / BENCHMARKING

NH-58 KM 3.25–3.50 end-to-end — same numbers in UI, tools, and work order.

Worked example · CRITICAL · RPI 86

- Inputs: 6.2 vs 2.8 mm/mo · 2.2× accel · 29.7/40 mm progress · 213 mm rain · crit 0.95 · conf 0.97
- kin = 100×(.4·0.62 + .3·0.40 + .3·0.74) = 59.4
- hazard = 59.4 × (1 + 0.5×1.0) rain-capped = 89.1
- RPI = 89.1 × 0.975 × 0.985 ≈ 85.7 → CRITICAL (threshold approach)
- Outlook: 51 d const · 44 d accel · widened → 35–64 days if trend continues — NOT a failure date
- History: this site 61→86 RPI · portfolio action sites 3→7 over 16 recomputed passes
- What-if: minor road → 57 ELEVATED · poor radar → ≤15 WEAK · rain on stable → ≈1 STABLE



Live Cloud Run + BigQuery (warm)

summary ~0.4 s · sites ~0.3 s · alerts ~2 s

Hot-path prefetch before history warm

Integrity: GenAI never computes RPI/tier/ETA

ADDITIONAL DETAILS / FUTURE DEVELOPMENT

Engineering invariants stay fixed. Scale the data plane — not the honesty rules.

Roadmap

- Production InSAR providers on the same Data Hub schema
- Materialize point-assignment for national corridors
- Multi-region HA once session store leaves single-process files
- Field checklist / offline packet for crews
- OIDC / IAP sign-in for enterprise buyers
- Deeper rain/geology as amplifiers only

Buyer · ask · honesty

- Buyer: asset-integrity / geotech teams (highway, metro, dam, mine)
- What they buy: engineer attention — 1,355 points → 7 sites
- Ask: live demo + 3-min video + repository
- Outlook bands ≠ collapse dates — always
- Gemini narrates tools only — always
- Synthetic demo data labelled in UI — always

SUBMISSION LINKS

GitHub · 3-minute demo video · live product.

GitHub public repository

https://github.com/ramasamyp-aiengineer/terra-sentinel-ai

Demo video (3 minutes)

docs/pitch/TerraSentinel-AI-Demo-3min.mp4 — upload to Drive/YouTube for the portal

Final product (live)

https://terrasentinel-598899050820.asia-southeast1.run.app

Sign-in: username + password are shown on the login page (judge demo hint). Judge walk: docs/JUDGE.md

RISK ENGINE · DETERMINISTIC

Measure → Score → Project. Plain rules first, then the exact maths.

1 · Measure

- Recent speed: slope last 60 d (mm/mo)
- Baseline: 270→90 d ago
- Accel = |recent| / max(|base|, 0.5)
- Persistence · seasonal recovery
- Site series = median in 250 m bin
- conf = .30coh+.15ep+.20pts+.20agr+.15SNR
- Weak data capped — never CRITICAL

2 · Score

- kin = 100·(.4 speed + .3 accel + .3 prog)
- hazard = kin·(1 + 0.5·rain14d/200)
- RPI = hazard × (0.5+0.5·crit)
- × (0.5+0.5·conf)
- Slow/far: ×0.2 · Seasonal: ×0.35
- WEAK cap 15 · quakes never scored
- CRITICAL / ELEVATED / WATCH rules

3 · Project

- ETA_const = remaining ÷ speed
- ETA_accel = root of $a \cdot t^2 + b \cdot t + c = T$
- band = [min,max] ≤ 365 d
- close? → [.75, 1.35]·mid
- Highway 40 mm/180 d · creep 2.0
- Metro 25 mm · Dam 30 mm (demo)
- ALWAYS a band — NEVER a date

TerraSentinel AI

Thank you — Team Agent Ram

Which 250 metres should an engineer inspect today?

The engine ranks them. Gemini writes the ticket. Every number is checked.

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Live demo + GitHub + 3-min video · Google Cloud AI Builder Cup 2026 · Sustainability

Gemini · Vertex AI · BigQuery · Cloud Run · Sentinel-1 InSAR

